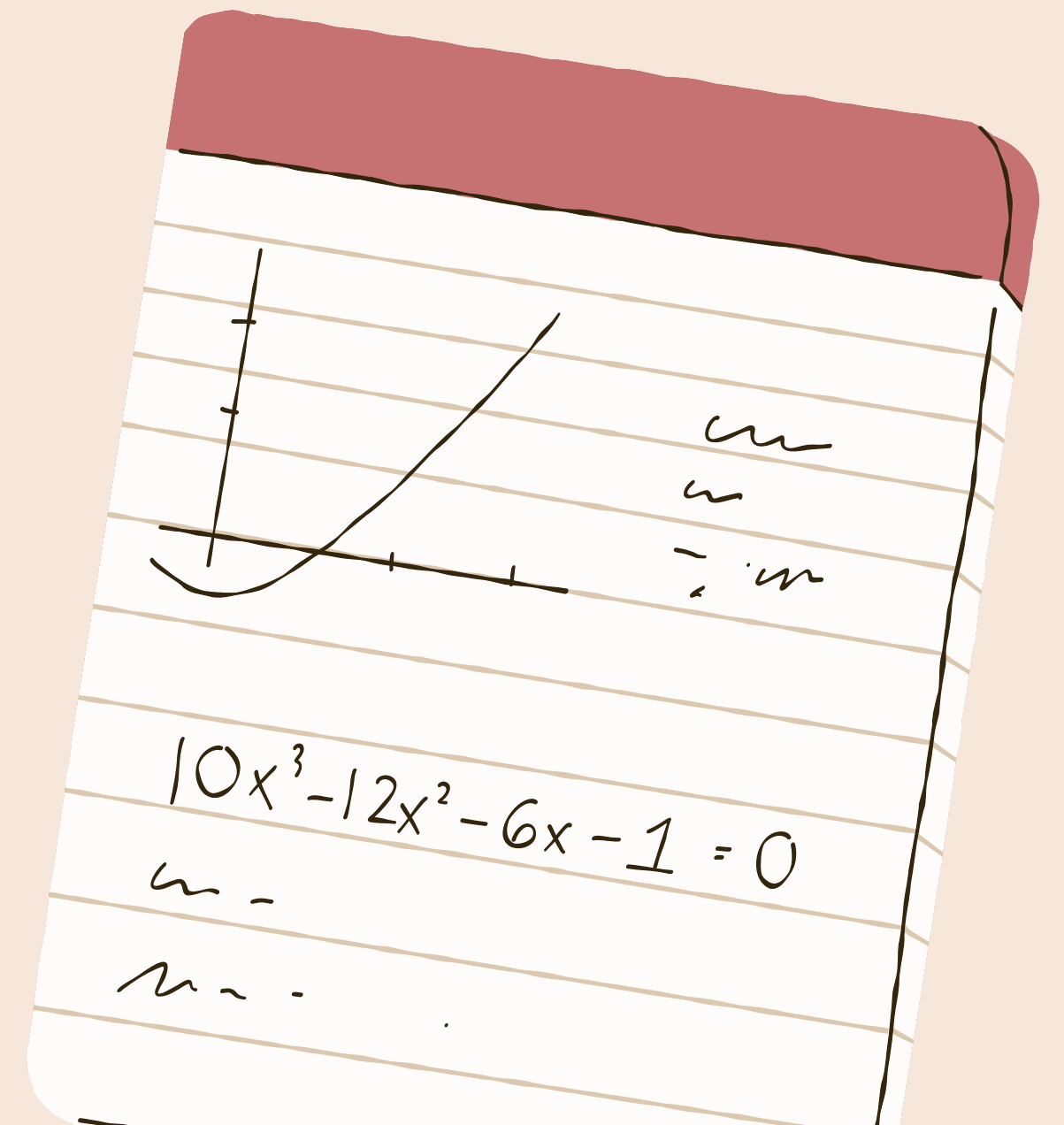
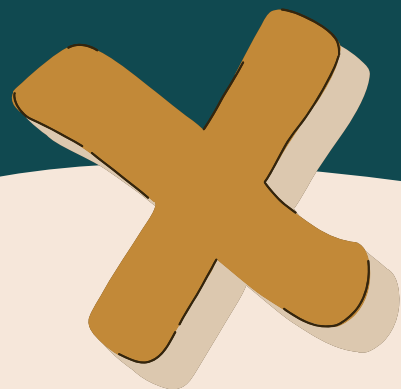
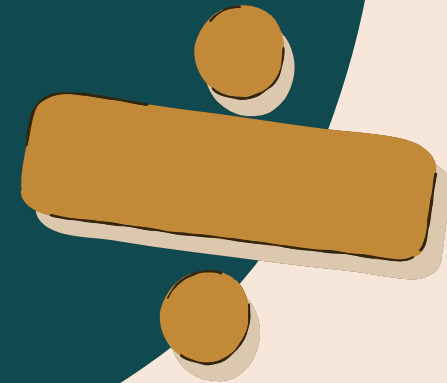
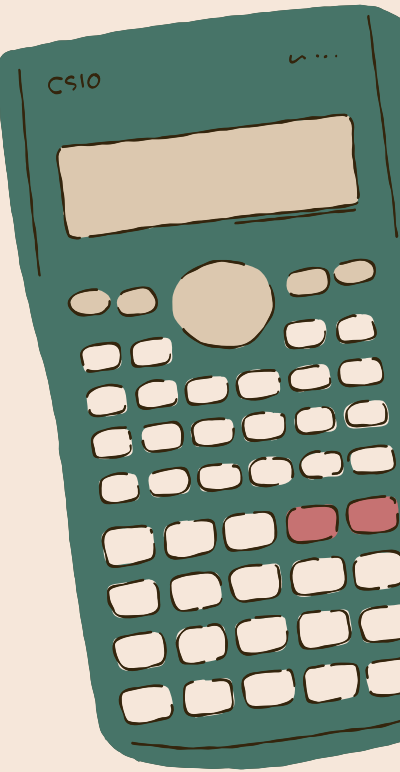


Year 7

Algebra



Variables and Operations

Variables:

- Letters representing values that change (e.g., x, y, n).

Constants:

- Numbers with fixed values (e.g., 9, 16).

Coefficients:

- A constant multiplying a variable (e.g., 3 in 3x).

(x)

(x)

Key phrases:

"Times," "product" → Multiplication

"Divided by," "quotient" → Division

"Sum," "plus" → Addition

"Difference," "minus" → Subtraction



Basic Operations in Algebra

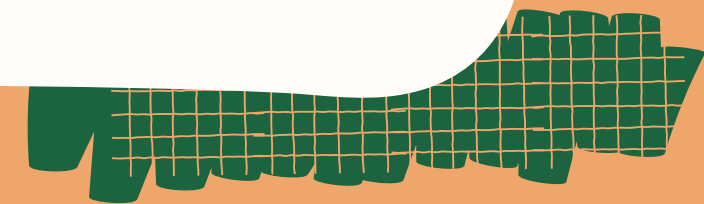
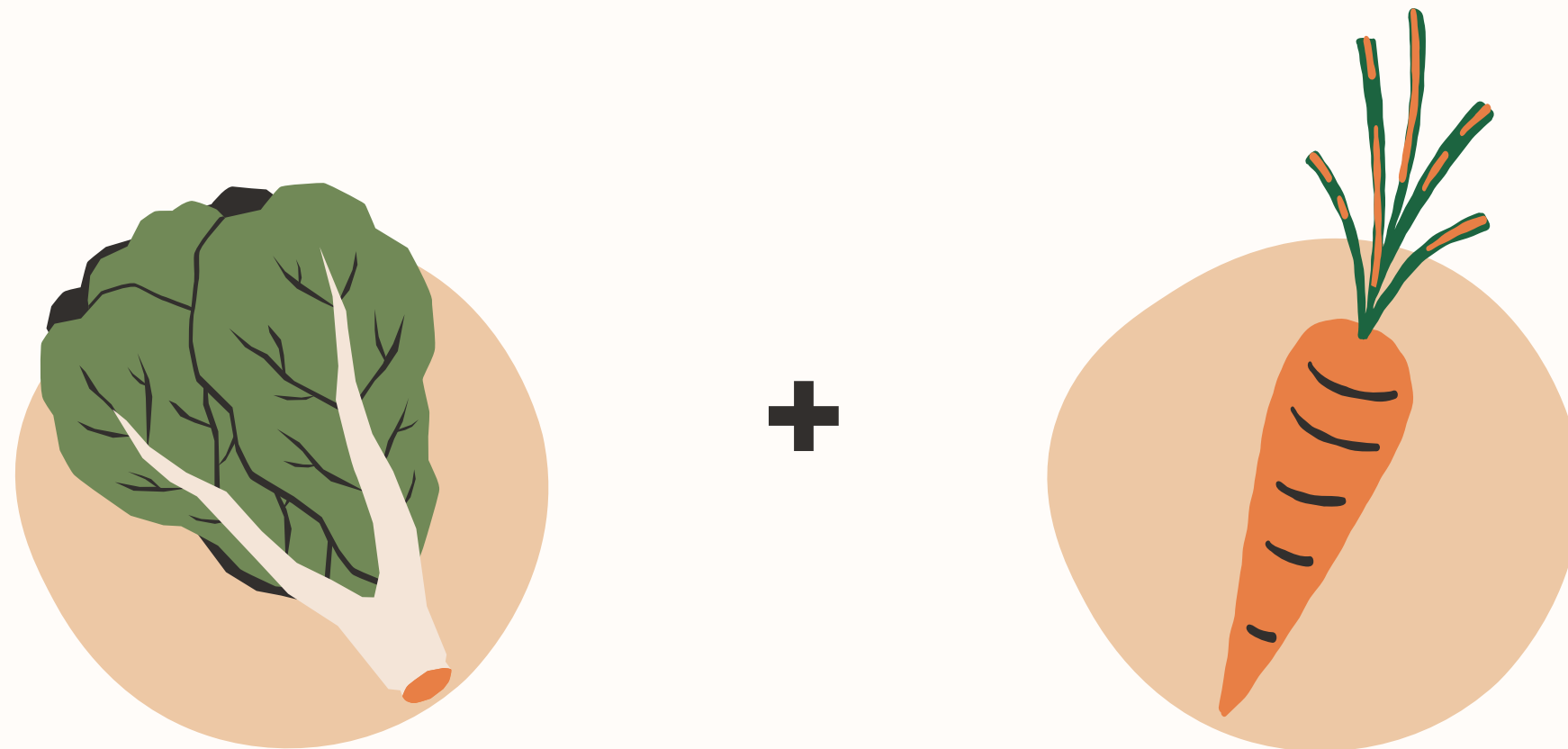
Addition, subtraction, multiplication and division are some of the operations used to create algebraic expressions.



Word Expression	Algebraic Expression	Simplified Expression
a increased by 5	$a + 5$	$a + 5$
2 less than a	$a - 2$	$a - 2$
a groups of b	$a \times b$	ab
a divided into b equal parts	$a \div b$	$\frac{a}{b}$



Imagine getting a head of lettuce and a piece of carrot in a supermarket.
What do you think is its total weight?

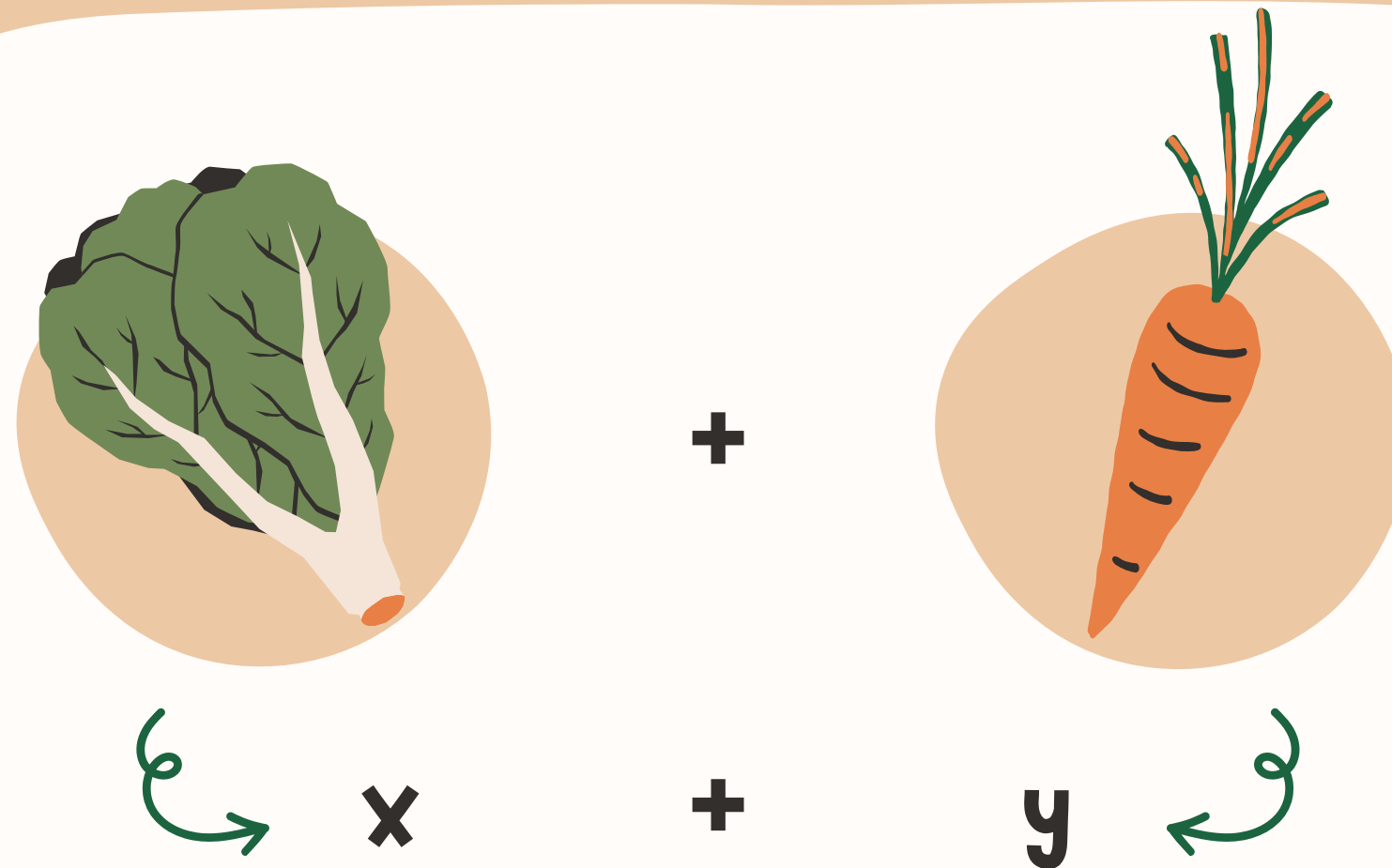




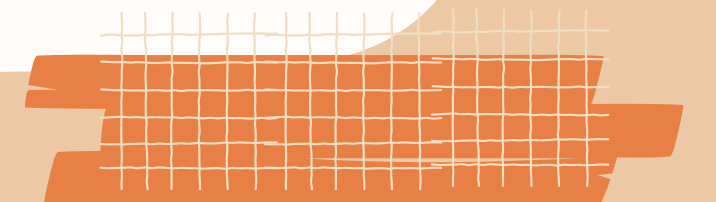
Creating Algebraic Expressions

Even if we don't know the actual weight of each vegetable, we can express its weight using algebraic expressions.

We can use variables or pronumerals in place of a numeric value. It can be a letter or a symbol.



In this case, x and y replace the weights of lettuce and carrot.



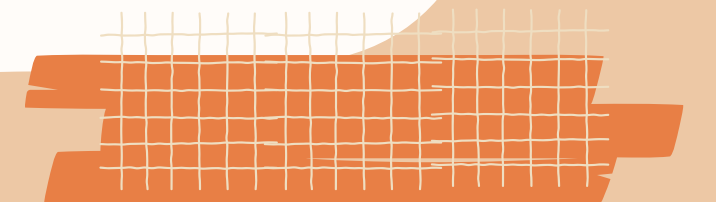
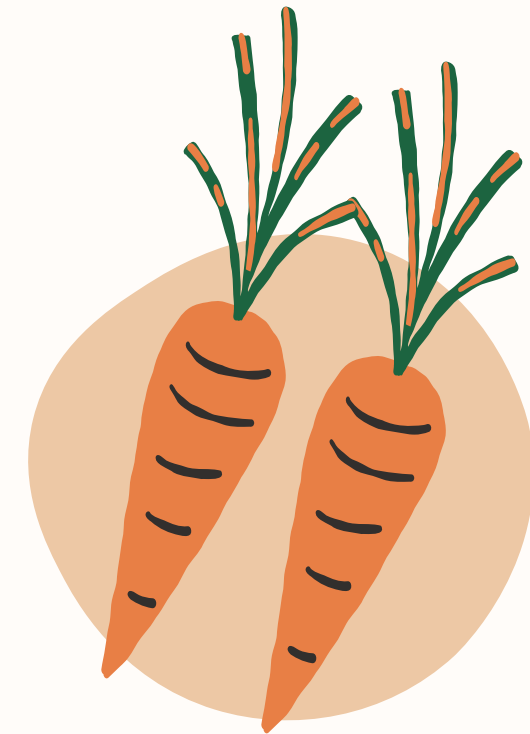


Creating Algebraic Expressions

What if we add another piece of carrot that has the same weight? How will the expression change?



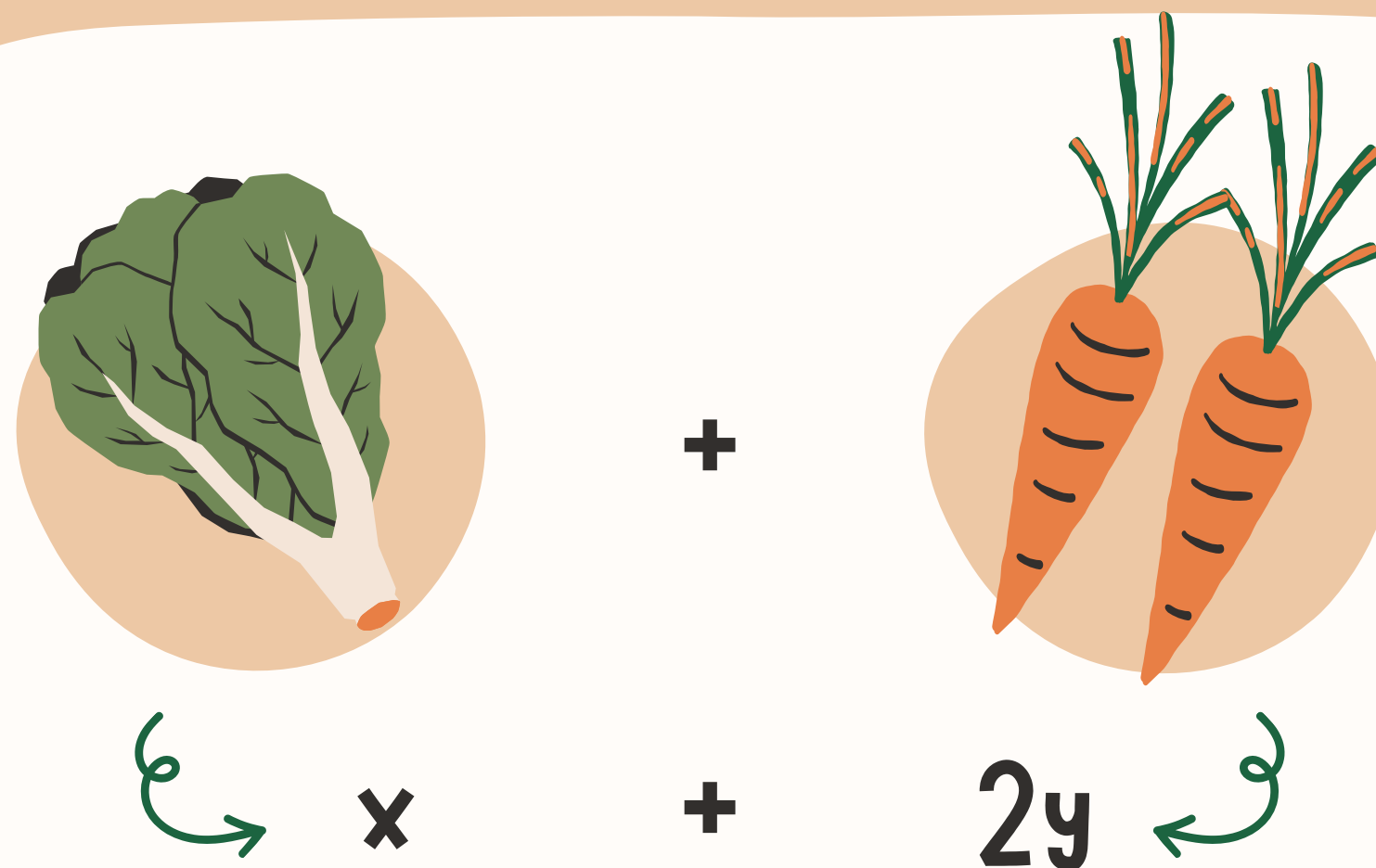
+



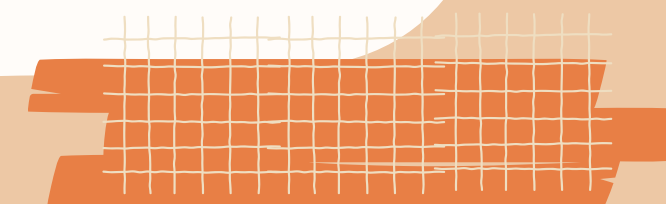


Creating Algebraic Expressions

This is where coefficients will come in. Coefficients are used in algebraic expressions to represent how many sets of a pronumeral we have.



In this case, 2 is a coefficient of y.



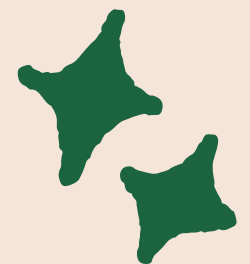
How can you express the total number of oranges?

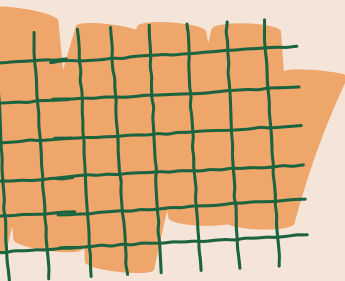


$$3b + 5c$$



The total number of oranges can be expressed as $3b + 5c$ where b represents the number of oranges in each basket and c represents the number of oranges in each cup.





How can you express the
total number of oranges?



Each basket has the same
number of oranges.



Each cup has the same
number of oranges.



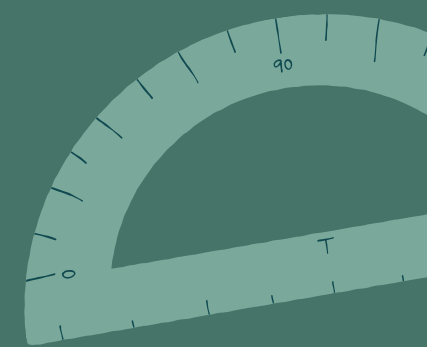
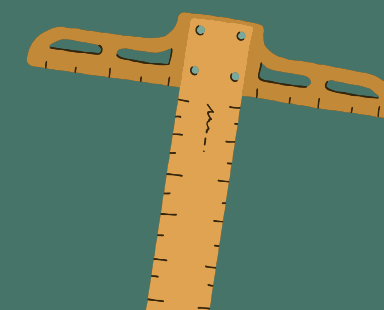


In algebra, the multiplication sign is omitted, and numbers are placed in front of letters.



Example

$$3 \times a$$
$$3a$$



Example

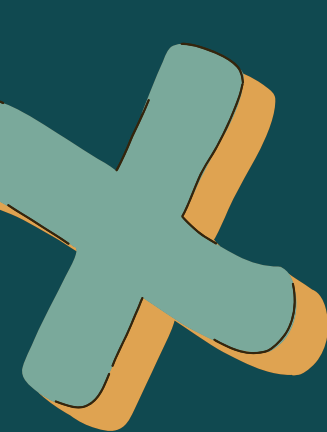
$$a \times b$$

$$ab$$

Example

$$p \times q \times r$$

$$pqr$$



Example

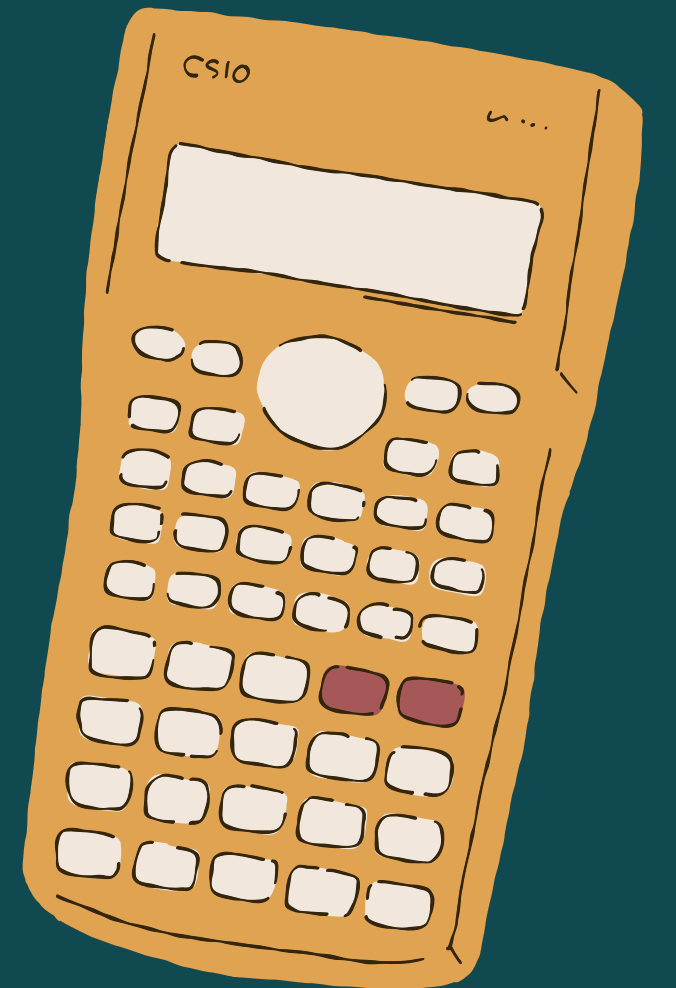
$$2a \times 4b$$

$$2 \times a \times 4 \times b$$

$$2 \times 4 \times a \times b$$

$$8ab$$


1. Multiply the numbers
2. Multiply the variables



Example

$$a \times a$$

$$a^2$$

Example

$$a \times a \times a$$

$$a^3$$



Expressions do not include an equal to sign or inequality sign.

Example: $23x - 46$

Equations always include an equal sign.

Example: $23x = 69$

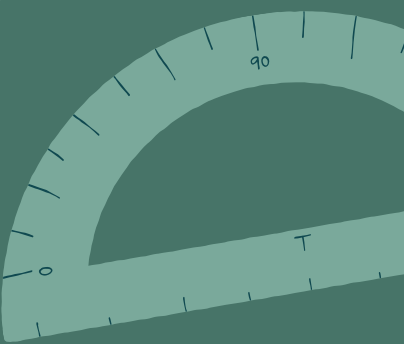
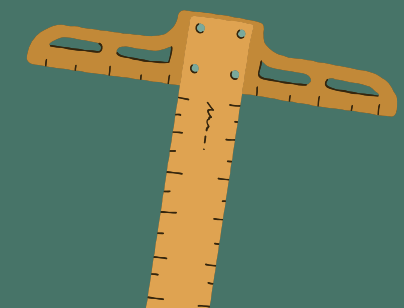
Expressions into words

Three more than a number $\rightarrow x + 3$

Five times as many $\rightarrow 5 \times \text{variable}$

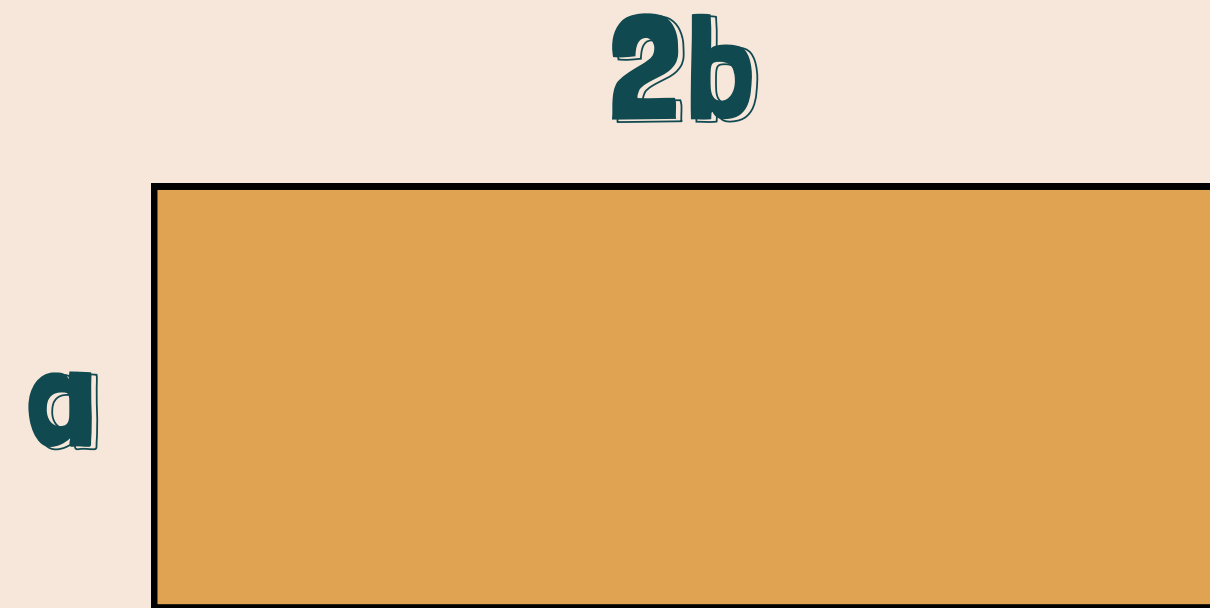
The sum of two numbers $\rightarrow x + y$

Percent of $\rightarrow \text{Decimal} \times \text{variable}$

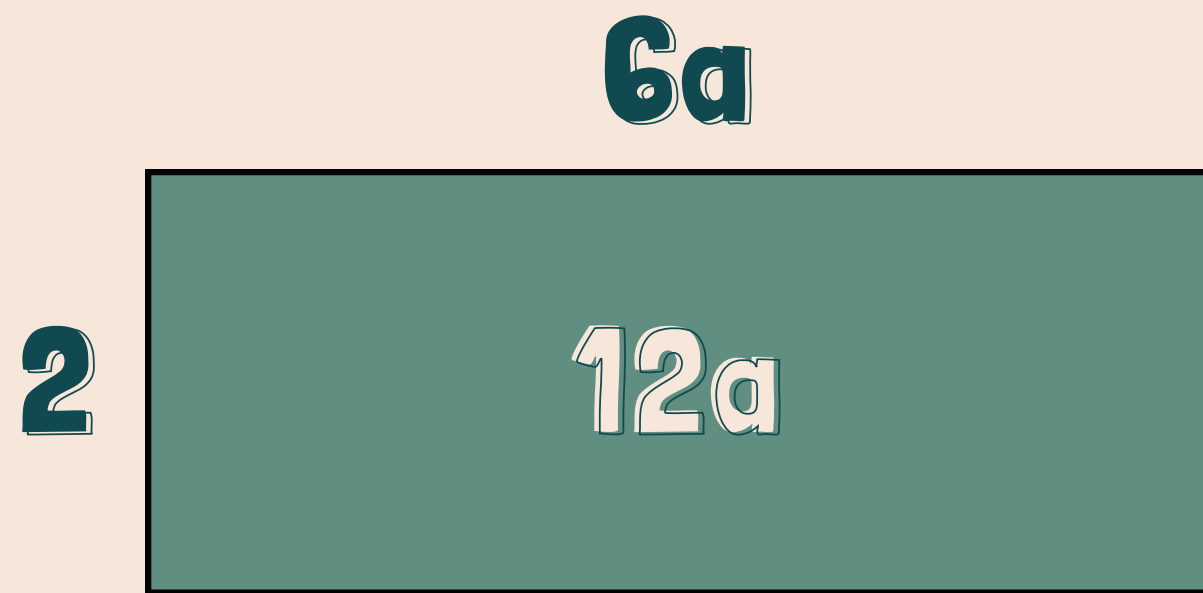


Multiplying Terms using the Area Model

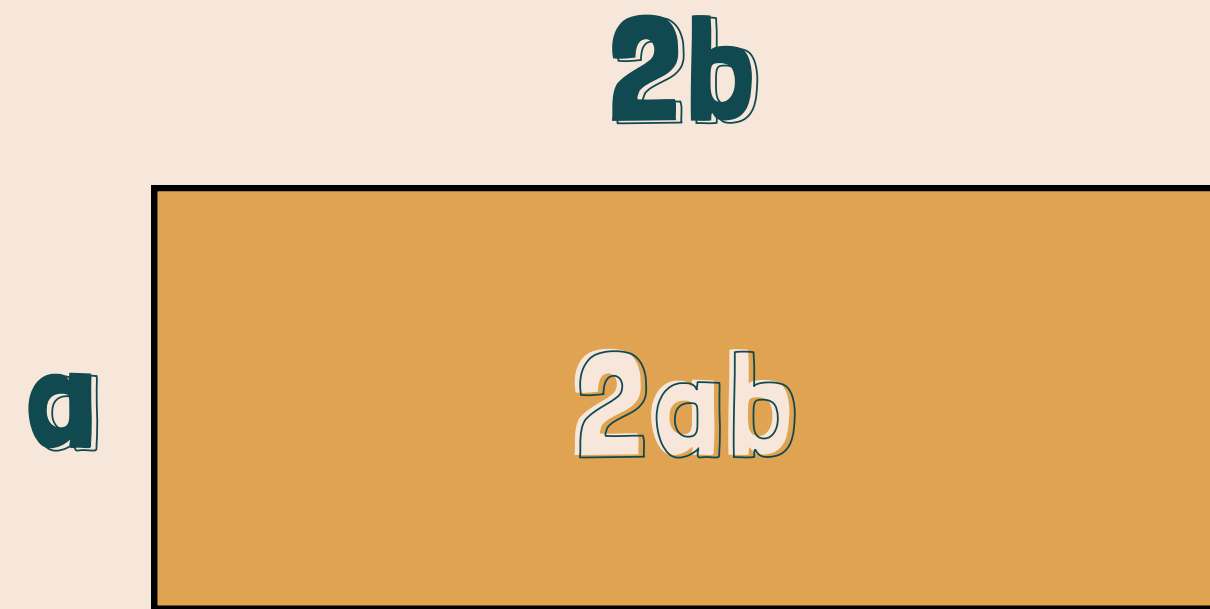
Can you figure out an expression for the area of each rectangle?



Multiplying Terms using the Area Model



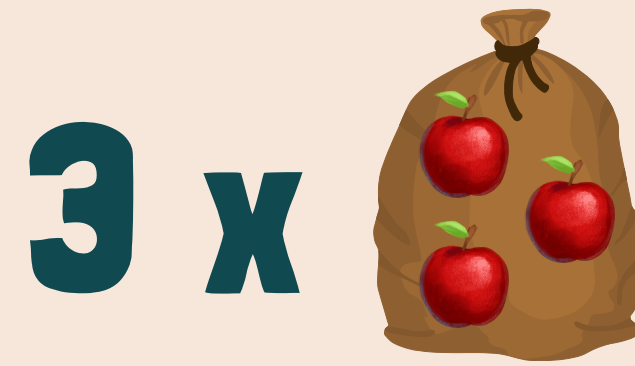
$$2 \times 6a = 12a$$



$$a \times 2b = 2ab$$

Simplifying Terms

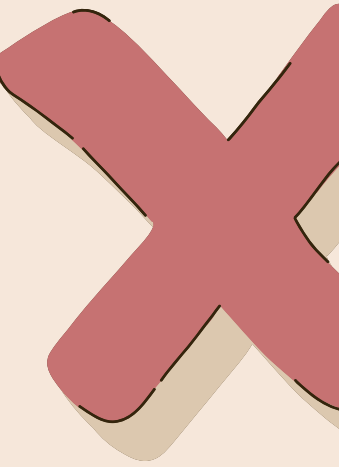
$y + y + y$ is the same as 3 multiplied by y .



$$y + y + y$$

$$3 \times y$$

Have a go...



$$2a \times 3$$

$$4a \times 6b$$

$$6a \times b$$

$$3a \times a$$

$$7a \times 4b$$

$$7a \times 2a$$

Answer Key

$$2a \times 3 = 6a$$

$$4a \times 6b = 24ab$$

$$6a \times b = 6ab$$

$$3a \times a = 3a^2$$

$$7a \times 4b = 28ab$$

$$7a \times 2a = 14a^2$$

Multiplying Terms with Indices

$$2a^2b \times 3ab$$

$$\boxed{2} \times \boxed{a} \times \boxed{a} \times \boxed{b} \times \boxed{3} \times \boxed{a} \times \boxed{b}$$

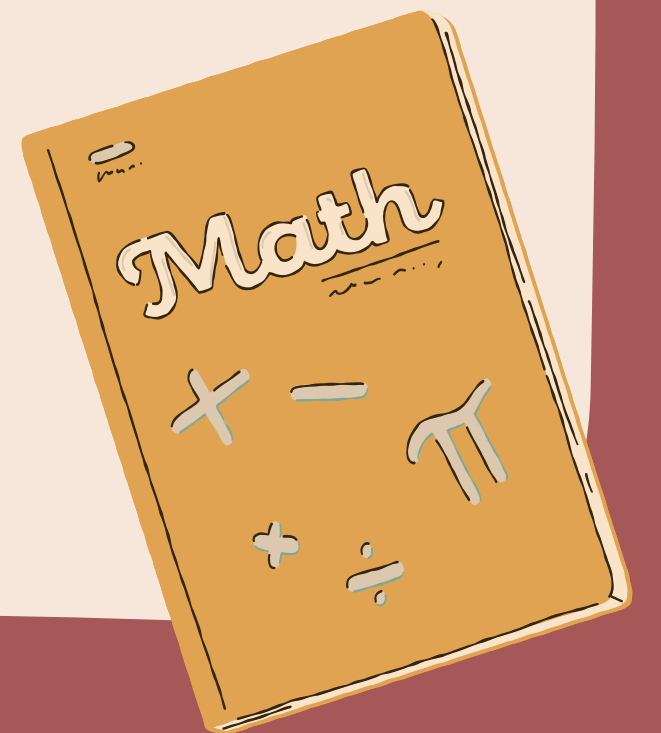
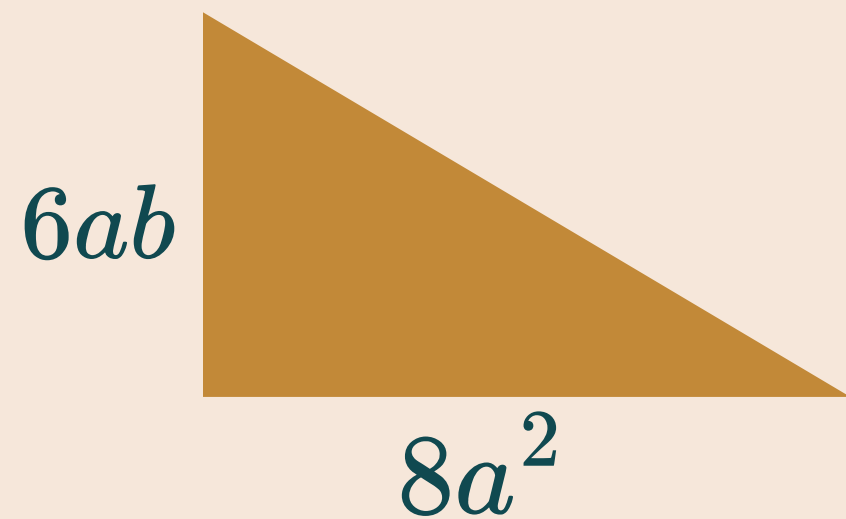
$$= \boxed{6} \times \boxed{a^3} \times \boxed{b^2}$$

$$= 6a^3b^2$$

Ready for a challenge?

1. Simplify the expression $4ab^2 \times 0.5a^2$

2. Find an expression for the area of the triangle

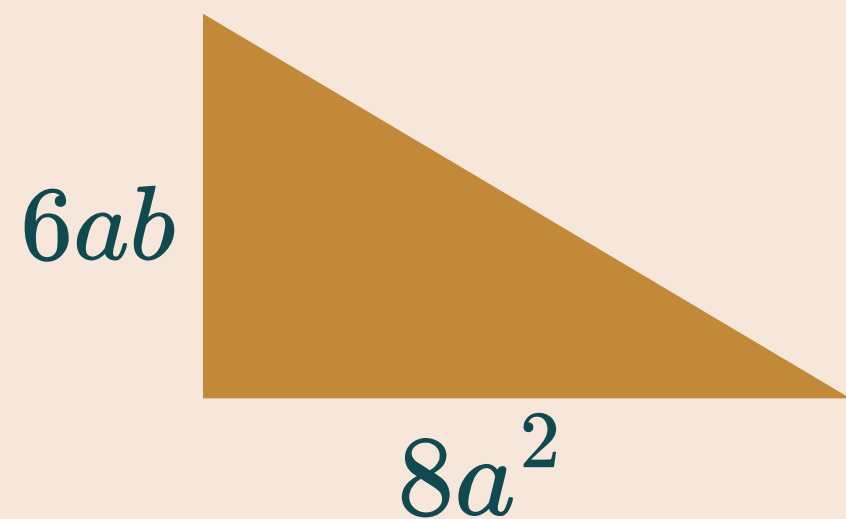


Solutions

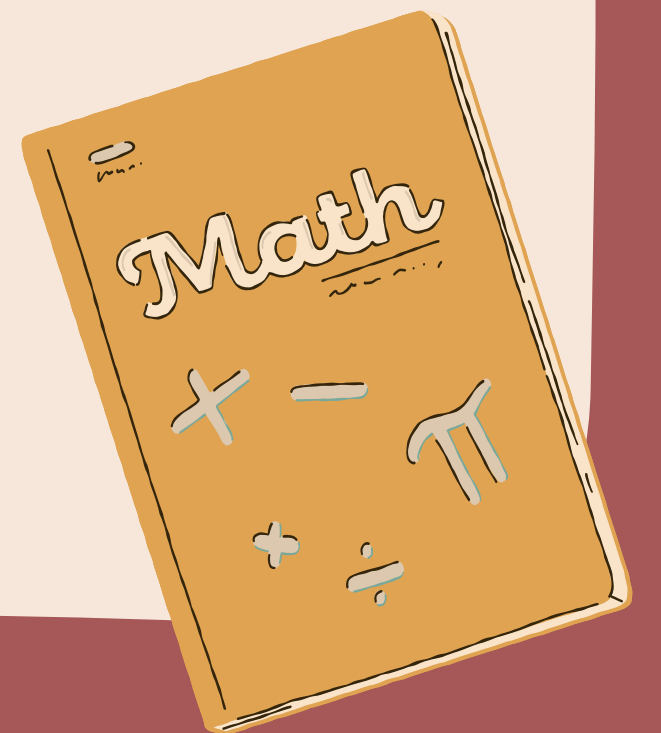
1. Simplify the expression $4ab^2 \times 0.5a^2$

$$= 2a^3b^2$$

2. Find an expression for the area of the triangle

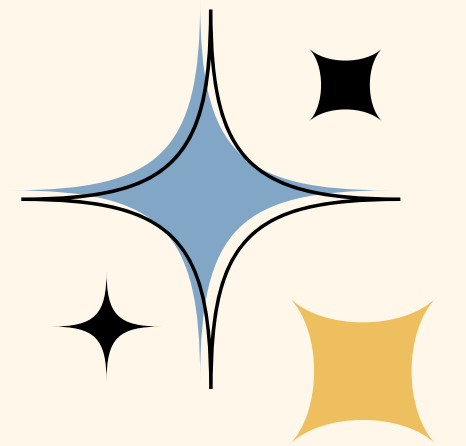


$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 6ab \times 8a^2 \\ \text{Area} &= 24a^3b \end{aligned}$$



1 Solving One Step Equations

Examples



$$x + 7 = 15$$

$$\quad -7 \quad -7$$

$$x = 8$$

$$3x = 15$$

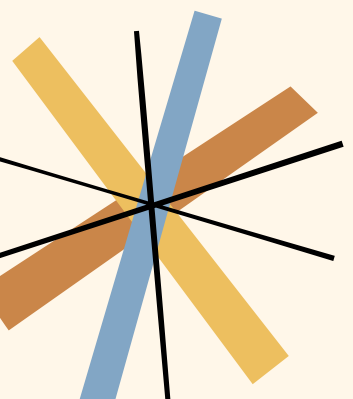
$$\div 3 \quad \div 3$$

$$x = 5$$

$$\frac{x}{4} = 3$$

$$\times 4 \quad \times 4$$

$$x = 12$$



Practice Questions

1

$$x + 9 = 13$$

2

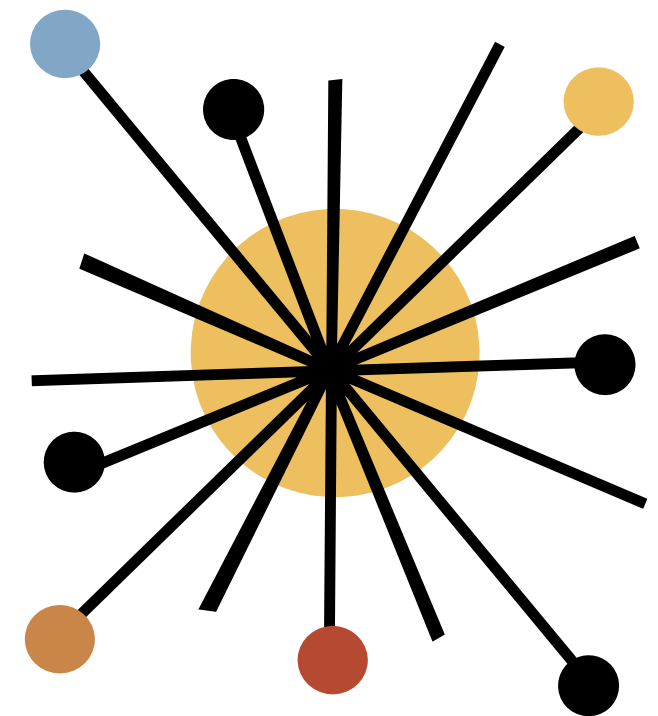
$$5 - x = 12$$

3

$$8x = 60$$

4

$$\frac{x}{6} = 12$$



Answers

1 $x + 9 = 13$

$x = 4$

2 $5 - x = 12$

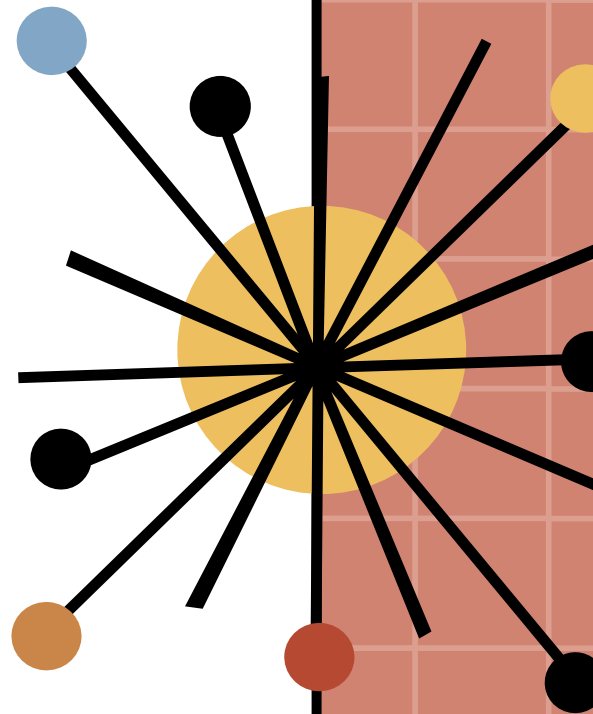
$x = -7$

3 $8x = 60$

$x = 7.5$

4 $\frac{x}{6} = 12$

$x = 72$



2 Solving Two Step Equations

Examples

$$2x - 3 = 15$$

$$+3 \quad +3$$

$$2x = 18$$

$$\div 2 \quad \div 2$$

$$x = 9$$

$$\frac{x}{4} + 6 = 9$$

$$-6 \quad -6$$

$$\frac{x}{4} = 3$$

$$\times 4 \quad \times 4$$

$$x = 12$$

Cross multiplication

[https://youtube.com/shorts/91MF-PfaWRQ?
si=UhNgllFwqE-v2qW6](https://youtube.com/shorts/91MF-PfaWRQ?si=UhNgllFwqE-v2qW6)

Practice Questions

1

$$2x + 5 = 14$$

2

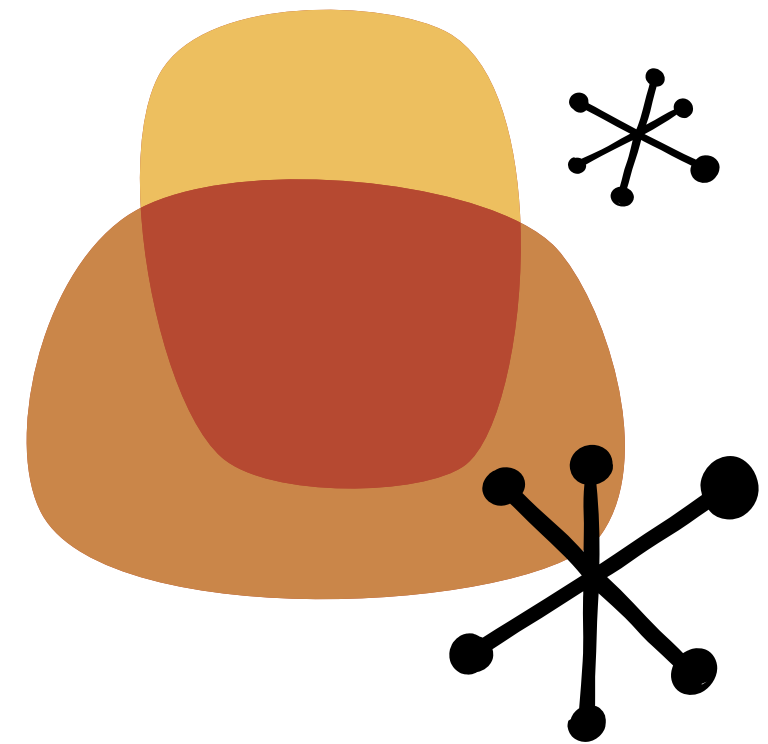
$$15 - 6x = 12$$

3

$$\frac{x}{3} - 2 = 7$$

4

$$\frac{x}{5} + 8 = 21$$



Answers

1 $2x + 5 = 14$

$x = 4.5$

2 $15 - 6x = 12$

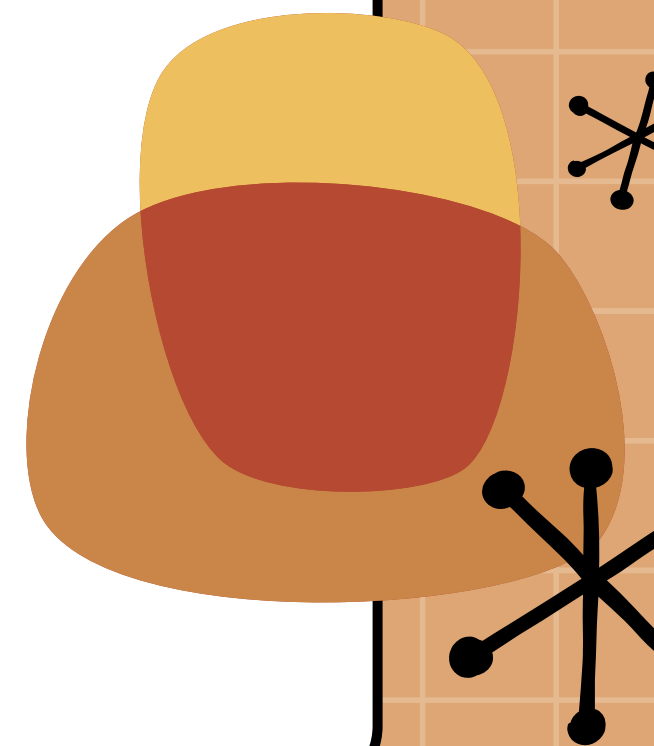
$x = 0.5$

3 $\frac{x}{3} - 2 = 7$

$x = 27$

4 $\frac{x}{5} + 8 = 21$

$x = 65$



Solving equations

Goal: Isolate the variable by itself on the other side of the equation.

Method: Use the opposite(inverse) operation on both sides of the equation.

Remember: What you do to one side, you must do to the other side!

Example:

$$x + 8 = 15$$

Subtract 8 from both sides, which
becomes $x = 7$

Example:

$$x/5 = 3$$

Multiply both sides by 5, we get $3(5) = 15$
 $x = 15$

$$S = 6L^2, L = 3$$

$$a = \frac{1}{2} gt^2, g = 10, t = 4$$

$$P = b^c, b = 10, c = 4$$

$$W = xy/z, x = 2, y = 6, z = 4$$

$$I = V/R, V = 12, R = 6$$

$$v = d/t, d = 42, t = 7$$

Thank you for listening

Any Questions?

